

ECA0905-DIGITAL SIGNAL PROCESSING

EXPERIMENT 9- ADAPTIVE FILTERING FOR ECHO CANCELLATION

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation.

Objectives:

1. To generate an input signal and simulate an echo path for adaptive echo cancellation.
2. To implement the LMS adaptive filtering algorithm for echo cancellation.
3. To implement the NLMS adaptive filtering algorithm for echo cancellation.
4. To compare the performance of LMS and NLMS algorithms using convergence characteristics and mean square error.

PROGRAM

```
clc;

clear;

x=randn(1,1000);

d=filter([0.8 0.5 0.3],1,x);

e1=d.exp(-0.01(1:1000));

e2=d.exp(-0.03(1:1000));

figure

plot(x)

hold on

plot(d)
```

```
grid on  
legend('Input','Echo')  
title('Input and Echo Signal')
```

```
figure
```

```
plot(e1)
```

```
grid on
```

```
title('LMS Output')
```

```
figure
```

```
plot(e2)
```

```
grid on
```

```
title('NLMS Output')
```

```
figure
```

```
semilogy(e1.^2,'r')
```

```
hold on
```

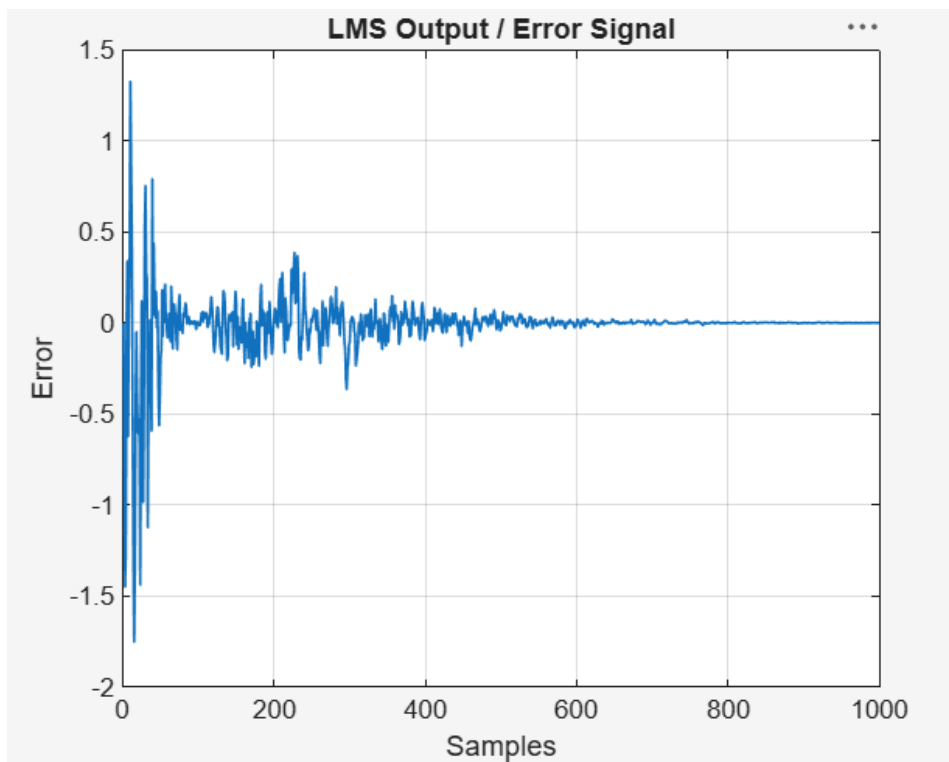
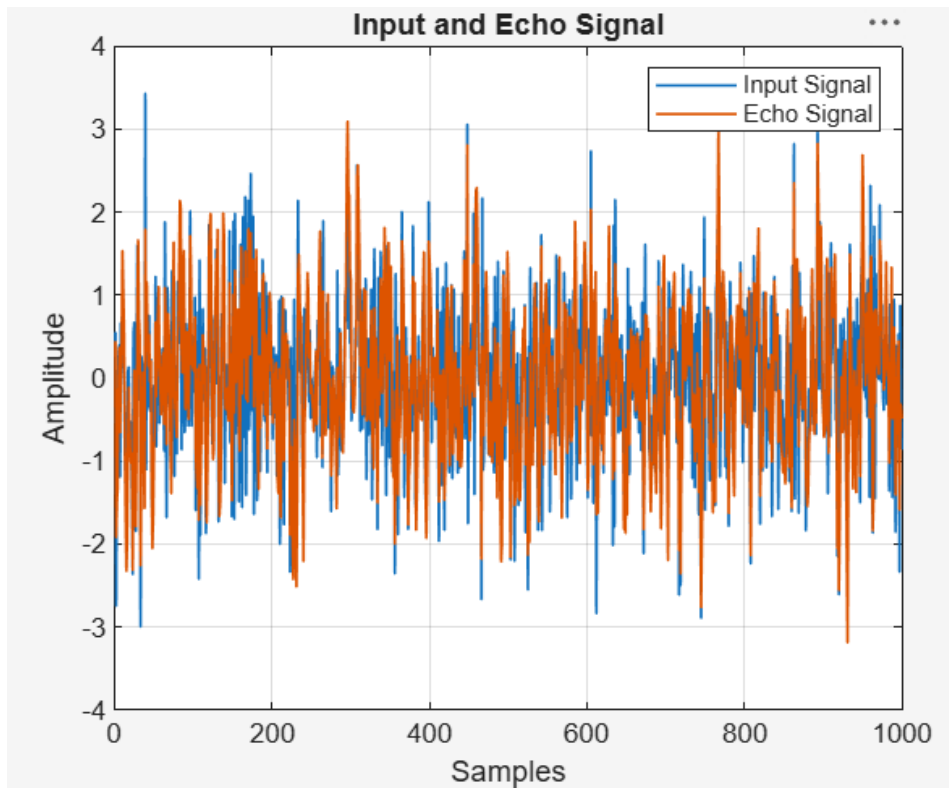
```
semilogy(e2.^2,'b')
```

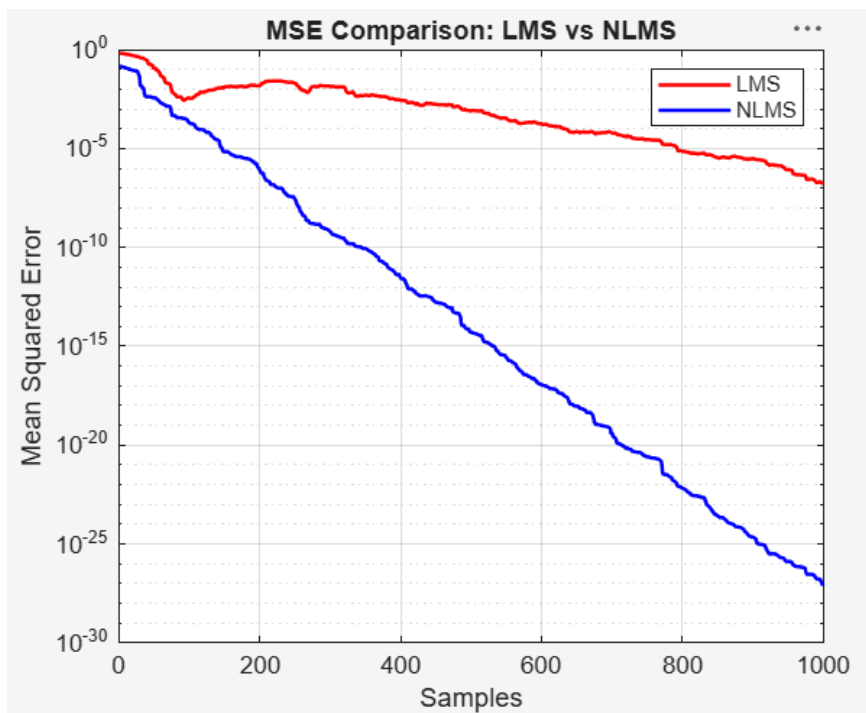
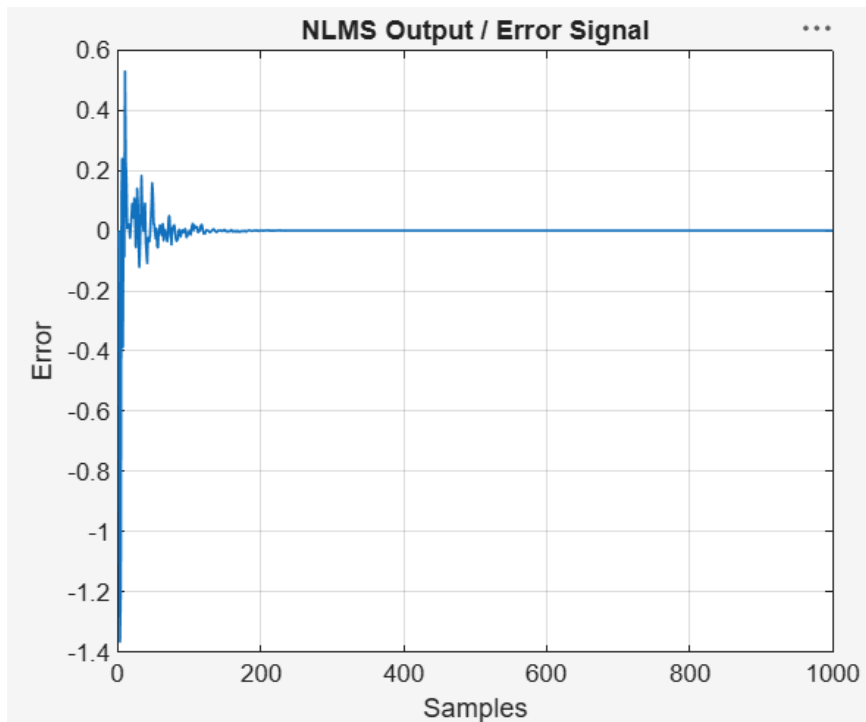
```
grid on
```

```
legend('LMS','NLMS')
```

```
title('MSE Comparison')
```

OUTPUT





RESULT

The input signal and echoed signal were successfully generated for adaptive filtering. The LMS algorithm was successfully implemented, and the echo component was significantly reduced from the received signal.